



Physics Worksheet

Conservation of Energy

Name: _____ Student #: _____ Student Email #: _____

1. Is the value of an object's mechanical energy meaningful, or is it only the difference in mechanical energy that is meaningful?
2. Aditi and Adam are having a disagreement about potential energy (as you do). Adam picked his (very heavy) 1 kg phone 0.5 m above a 0.5 m high coffee table. Adam states the phone has 9.8 J of potential energy, as where Aditi states it actually has 4.9 J. Which one of them is correct?
3. Derive the formula for potential energy expressed as mgh . (Hint: look up gravitational potential energy)

4. This question concerns the spring force
- (a) Is the spring force conservative? Explain with your own words why or why not *and* prove it mathematically

 - (b) Derive the potential energy for a spring. Justify why this is possible
5. You are tasked to find how powerful of a propelling engine you need for a satellite after it's initial launch. Your 1000kg satellite needs to reach Earth's escape velocity (11.0 km s^{-1}) between the heights of 50 km and 100 km above the Earth's surface starting from a Lagrangian of $9.9 \times 10^8 \text{ J}$. Ignoring air drag, how much power will your engine have to expend?